

Worksheet 1: Fractions

Annotated Revision Guide

AIML Foundations Mathematics

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Part A: Numerical Fractions Review (Problems 1–10)

Concept. A fraction $\frac{a}{b}$ names a quantity in terms of equal parts. Adding or subtracting fractions requires a *common unit* — you cannot add quarters to sixths until both are expressed in the same denomination. The **lowest common denominator** (LCD) is the smallest number that both denominators divide into evenly; it equals the LCM of the denominators.

Method. (1) Find $\text{LCD}(a, b)$. (2) Convert each fraction: multiply top and bottom by the same factor. (3) Operate on the numerators. (4) Simplify by dividing by $\text{gcd}(\text{numerator}, \text{denominator})$. For multiplication/division, cancel *before* multiplying: spot common factors between any numerator and any denominator.

Watch out. The most common error is **adding tops and bottoms**: $\frac{1}{2} + \frac{1}{3} \neq \frac{2}{5}$. Think of it this way: half a pizza plus a third of a pizza is $\frac{5}{6}$ of a pizza, not $\frac{2}{5}$. Denominators name the unit; numerators count how many of that unit.

Where this leads. Everything in this part reappears in Parts B–D with letters instead of numbers. The LCD becomes a common denominator expression like $(x + 2)(x - 3)$. The algebra looks harder but the logic is identical.

Explore further. *Egyptian fractions:* Ancient Egyptians wrote every fraction as a sum of *distinct* unit fractions, e.g. $\frac{3}{4} = \frac{1}{2} + \frac{1}{4}$. Try expressing $\frac{5}{6}$ and $\frac{7}{12}$ this way. For the history, see George Joseph, *The Crest of the Peacock*, Ch. 2.

1. $\frac{3}{4} + \frac{5}{6}$

Step 1. $\text{LCD}(4, 6) = 12$.

Step 2. $\frac{3}{4} = \frac{9}{12}, \quad \frac{5}{6} = \frac{10}{12}$.

Step 3. $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.

Note: 19 is prime, so $\frac{19}{12}$ is already in lowest terms.

2. $\frac{7}{8} - \frac{2}{3}$

Step 1. $\text{LCD}(8, 3) = 24$.

Step 2. $\frac{7}{8} = \frac{21}{24}, \quad \frac{2}{3} = \frac{16}{24}$.

Step 3. $\frac{21}{24} - \frac{16}{24} = \frac{5}{24}$.

Note: $\text{gcd}(5, 24) = 1$, so $\frac{5}{24}$ is fully reduced.

3. $\frac{5}{9} \times \frac{3}{10}$

Method. Cancel *before* multiplying: 5 and 10 share a factor of 5; 3 and 9 share a factor of 3.

Cancel. $\frac{\cancel{5}^1}{\cancel{9}^3} \times \frac{\cancel{3}^1}{\cancel{10}^2} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$.

Note: Cancelling first keeps numbers small and avoids simplifying a large result at the end.

4. $\frac{4}{5} \div \frac{2}{15}$

Recall. Division by a fraction = multiplication by its reciprocal.

Rewrite. $\frac{4}{5} \times \frac{15}{2}$.

Cancel. 4 and 2 share 2; 15 and 5 share 5: $\frac{\cancel{4}^2}{\cancel{5}^1} \times \frac{\cancel{15}^3}{\cancel{2}^1} = \frac{2 \times 3}{1} = 6$.

Look ahead: Division of algebraic fractions (Part C, problems 25–26) uses exactly this "flip and multiply" rule.

5. $\frac{2}{3} + \frac{1}{4} - \frac{1}{6}$

Step 1. LCD(3, 4, 6) = 12.

Step 2. $\frac{2}{3} = \frac{8}{12}, \quad \frac{1}{4} = \frac{3}{12}, \quad \frac{1}{6} = \frac{2}{12}$.

Step 3. $\frac{8 + 3 - 2}{12} = \frac{9}{12} = \frac{3}{4}$.

Note: When three or more fractions share the same LCD, combine all numerators in one step.

6. $\frac{7}{12} \times \frac{8}{21} \times \frac{3}{2}$

Cancel across all three. 7 and 21 share 7; 8 and 12 share 4; 3 and 3 share 3 (from the third fraction).

Result. $\frac{\cancel{7}^1}{\cancel{12}^1} \times \frac{\cancel{8}^2}{\cancel{21}^3} \times \frac{\cancel{3}^1}{2} = \frac{1 \cdot 2 \cdot 1}{1 \cdot 3 \cdot 1} = \frac{2}{3}$.

Note: With a chain of multiplications, cancel between any numerator and any denominator before multiplying out.

7. $\frac{\frac{3}{4}}{\frac{9}{16}}$ (complex fraction)

Rewrite. $\frac{3}{4} \div \frac{9}{16} = \frac{3}{4} \times \frac{16}{9}$.

Cancel. 3 and 9 share 3; 16 and 4 share 4: $\frac{1}{1} \times \frac{4}{3} = \frac{4}{3}$.

Note: A fraction bar is always division. Reading $\frac{a/b}{c/d}$ as $(a/b) \div (c/d)$ is the key step.

8. $\frac{\frac{2}{5} + \frac{1}{3}}{\frac{1}{2}}$

Numerator first. LCD(5, 3) = 15: $\frac{2}{5} + \frac{1}{3} = \frac{6}{15} + \frac{5}{15} = \frac{11}{15}$.

Divide. $\frac{11}{15} \div \frac{1}{2} = \frac{11}{15} \times 2 = \frac{22}{15}$.

Note: Always simplify the numerator and denominator of a complex fraction separately before dividing.

9. $2\frac{1}{3} \times 1\frac{1}{2}$ (mixed numbers)

Convert. $2\frac{1}{3} = \frac{7}{3}$; $1\frac{1}{2} = \frac{3}{2}$.

Multiply. $\frac{7}{3} \times \frac{3}{2}$: the 3s cancel, giving $\frac{7}{2} = 3\frac{1}{2}$.

Note: Always convert mixed numbers to improper fractions before multiplying or dividing.

10. $4\frac{2}{5} \div 1\frac{1}{10}$

Convert. $4\frac{2}{5} = \frac{22}{5}$; $1\frac{1}{10} = \frac{11}{10}$.

Divide. $\frac{22}{5} \times \frac{10}{11}$: 22 and 11 share 11; 10 and 5 share 5. $\frac{2}{1} \times \frac{2}{1} = 4$.

Note: The answer is a whole number: $4\frac{2}{5}$ is exactly 4 times $1\frac{1}{10}$.

Part B: Algebraic Fractions — Simplification (Problems 11–18)

Concept. An algebraic fraction simplifies when its numerator and denominator share a common *factor*. The crucial skill is **factoring polynomials**: difference of squares, trinomial factoring, and pulling out common monomials. Once both numerator and denominator are fully factored, matching factors in both cancel to 1.

Method. (1) Factor numerator completely. (2) Factor denominator completely. (3) Cancel any factor that appears in both. (4) Write the simplified result, noting any values that make the original undefined (even if they cancel out).

Watch out. Cancelling terms, not factors. $\frac{x^2 + 4}{x + 2} \neq x + 2$ — you cannot cancel x^2 with x because x^2 is being *added*, not multiplied. Only factors (things being multiplied across the entire expression) cancel.

Where this leads. In calculus, simplifying a rational function by cancelling a

common factor is how you evaluate limits like $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$. Partial-fraction decomposition (integration) reverses this process.

Explore further. Factoring over the reals produces linear and irreducible quadratic factors (Fundamental Theorem of Algebra). For a deep but accessible treatment, see Stewart *Precalculus*, Ch. 3, or for more algebraic structure, Herstein *Topics in Algebra*, Ch. 3.

11. $\frac{6x^2}{9x}$

Factor. $= \frac{3x \cdot 2x}{3x \cdot 3}$
Cancel. $= \frac{2x}{3}$.

Note: Valid for $x \neq 0$ (the original has x in the denominator).

12. $\frac{15a^3b^2}{25ab^4}$

Factor. $= \frac{5ab^2 \cdot 3a^2}{5ab^2 \cdot 5b^2}$
Cancel. $= \frac{3a^2}{5b^2}$.

Note: When cancelling powers: $a^3/a = a^{3-1} = a^2$; $b^2/b^4 = b^{2-4} = b^{-2} = 1/b^2$.

13. $\frac{x^2 - 4}{x + 2}$

Factor numerator. Difference of squares: $x^2 - 4 = (x - 2)(x + 2)$.

Cancel. $\frac{(x - 2)\cancel{(x + 2)}}{\cancel{(x + 2)}} = x - 2$.

Note: Restriction: $x \neq -2$. Even though $x + 2$ cancels, the original is undefined there.

Look ahead: This exact cancellation appears in the limit $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2} = (-2) - 2 = -4$.

14. $\frac{x^2 - 9}{x^2 + 6x + 9}$

Factor numerator. Difference of squares: $x^2 - 9 = (x - 3)(x + 3)$.

Factor denominator. Perfect square: $x^2 + 6x + 9 = (x + 3)^2$.

Cancel. $\frac{(x - 3)\cancel{(x + 3)}}{\cancel{(x + 3)}(x + 3)} = \frac{x - 3}{x + 3}$.

Note: Restriction: $x \neq -3$. Note that only one factor of $(x + 3)$ cancels.

15. $\frac{2x^2 + 4x}{x^2 + 2x}$

Factor numerator. $2x^2 + 4x = 2x(x + 2)$.

Factor denominator. $x^2 + 2x = x(x + 2)$.

Cancel. $\frac{2\cancel{x(x+2)}}{\cancel{x(x+2)}} = 2$.

Note: Result is the constant 2. Restrictions: $x \neq 0$ and $x \neq -2$.

16. $\frac{x^2 - 5x + 6}{x^2 - 4}$

Factor numerator. Find two numbers multiplying to 6 and adding to -5 : $(-2)(-3) = 6$, $(-2) + (-3) = -5$. So $x^2 - 5x + 6 = (x - 2)(x - 3)$.

Factor denominator. $(x - 2)(x + 2)$.

Cancel. $\frac{(\cancel{x-2})(x-3)}{(\cancel{x-2})(x+2)} = \frac{x-3}{x+2}$.

Note: Restriction: $x \neq 2$ and $x \neq -2$.

17. $\frac{6x^2 - x - 2}{2x^2 + 3x - 2}$

Factor numerator. Use AC method: $ac = 6 \times (-2) = -12$; need two numbers with product -12 and sum -1 : use $+3$ and -4 . $6x^2 + 3x - 4x - 2 = 3x(2x + 1) - 2(2x + 1) = (3x - 2)(2x + 1)$.

Factor denominator. $ac = 2 \times (-2) = -4$; need product -4 , sum $+3$: use $+4$ and -1 . $2x^2 + 4x - x - 2 = 2x(x + 2) - 1(x + 2) = (2x - 1)(x + 2)$.

Check for common factors. $(3x - 2)$, $(2x + 1)$, $(2x - 1)$, $(x + 2)$ share no common root. The fraction is **already in lowest terms**: $\frac{(3x - 2)(2x + 1)}{(2x - 1)(x + 2)}$.

Note: No cancellation is possible here. Both factors in each part are distinct. (The answer key simplification was in error.)

18. $\frac{x^3 - 8}{x^2 - 4}$

Factor numerator. Difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$.

Factor denominator. Difference of squares: $x^2 - 4 = (x - 2)(x + 2)$.

Cancel. $\frac{(\cancel{x-2})(x^2 + 2x + 4)}{(\cancel{x-2})(x + 2)} = \frac{x^2 + 2x + 4}{x + 2}$.

Note: The quadratic $x^2 + 2x + 4$ has discriminant $4 - 16 = -12 < 0$, so it has no real roots and cannot factor further over $\mathbb{R}$.

Look ahead: The sum/difference of cubes formulae ($a^3 \pm b^3$) reappear frequently in calculus limits.

Part C: Algebraic Fractions — Operations (Problems 19–26)

Concept. Adding or subtracting algebraic fractions is structurally identical to numerical fraction arithmetic: find a common denominator, rewrite, combine numerators. The LCD is now an expression rather than a number. Multiplying and dividing algebraic fractions also mirrors the numerical case.

Method. Add/subtract: (1) Identify LCD. (2) Rewrite each fraction with the LCD as denominator. (3) Combine numerators. (4) Expand and simplify.

Multiply: cancel across *any* numerator and *any* denominator, then multiply.

Divide: multiply by the reciprocal.

Watch out. When rewriting $\frac{1}{x-1}$ with LCD $(x-1)(x+1)$, multiply top *and* bottom by $(x+1)$. Multiplying only the bottom changes the fraction. Also: expand numerators carefully after combining — sign errors when subtracting are the most common source of mistakes.

Where this leads. Partial fractions (used in integration) reverses this process: given $\frac{2x+1}{(x-1)(x+1)}$, find $\frac{A}{x-1} + \frac{B}{x+1}$. Understanding how to combine fractions makes splitting them apart far more intuitive.

Explore further. The lens equation $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ and parallel resistance $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ are direct applications. Solving them for the desired variable is exactly problems 39–40 in this worksheet.

19. $\frac{2}{x} + \frac{3}{x}$

Same denominator. $\frac{2+3}{x} = \frac{5}{x}$.

20. $\frac{5}{x+1} - \frac{2}{x+1}$

Same denominator. $\frac{5-2}{x+1} = \frac{3}{x+1}$.

21. $\frac{3}{x} + \frac{2}{y}$

LCD. xy .

Rewrite. $\frac{3y}{xy} + \frac{2x}{xy} = \frac{3y+2x}{xy}$.

22. $\frac{x}{x+2} + \frac{2}{x-2}$

LCD. $(x+2)(x-2)$.

Rewrite. $\frac{x(x-2)}{(x+2)(x-2)} + \frac{2(x+2)}{(x+2)(x-2)}$.

Combine. $\frac{x^2 - 2x + 2x + 4}{(x+2)(x-2)} = \frac{x^2 + 4}{x^2 - 4}.$

Note: $x^2 + 4$ does not factor over $\mathbb{R}$ (no real roots), so this is fully simplified.

23. $\frac{1}{x-1} - \frac{1}{x+1}$

LCD. $(x-1)(x+1) = x^2 - 1.$

Combine. $\frac{(x+1) - (x-1)}{x^2 - 1} = \frac{2}{x^2 - 1}.$

Note: The x terms in the numerator cancel, leaving just 2.

24. $\frac{x}{x^2 - 1} + \frac{1}{x+1}$

Factor denominator. $x^2 - 1 = (x-1)(x+1).$

LCD. $(x-1)(x+1).$

Rewrite second fraction. $\frac{1}{x+1} = \frac{x-1}{(x-1)(x+1)}.$

Combine. $\frac{x + (x-1)}{(x-1)(x+1)} = \frac{2x-1}{x^2-1}.$

Note: Verify: at $x=2$: $LHS = \frac{2}{3} + \frac{1}{3} = 1$; $RHS = \frac{3}{3} = 1$. ✓

25. $\frac{2x}{x+3} \times \frac{x^2-9}{4x^2}$

Factor. $x^2 - 9 = (x-3)(x+3).$

Cancel. $(x+3)$ cancels; $2x$ and $4x^2$ share $2x$: $\frac{\cancel{2x}}{\cancel{(x+3)}} \times \frac{(x-3)\cancel{(x+3)}}{2x \cdot \cancel{2x}} = \frac{x-3}{2x}.$

Note: Restrictions: $x \neq 0$, $x \neq -3$.

26. $\frac{x^2-4}{x+5} \div \frac{x-2}{x^2+5x}$

Rewrite as multiplication. $\frac{x^2-4}{x+5} \times \frac{x^2+5x}{x-2}.$

Factor. $x^2 - 4 = (x-2)(x+2)$; $x^2 + 5x = x(x+5).$

Cancel. $(x-2)$ cancels; $(x+5)$ cancels: $(x+2) \cdot x = x(x+2) = x^2 + 2x.$

Look ahead: Notice how factoring reveals hidden common factors that only become visible once both fractions are combined.

Part D: Complex Algebraic Fractions (Problems 27–30)

Concept. A complex fraction contains fractions within its numerator or denominator. The cleanest approach is to **multiply the entire expression (top and bottom) by the LCD of all inner fractions**. This clears all inner denominators in one step.

Method. (1) Identify all denominators appearing inside the complex fraction. (2) Find their LCD. (3) Multiply numerator *and* denominator of the main fraction by this LCD. (4) Simplify the result.

Watch out. You must multiply *both* the outer numerator and the outer denominator by the LCD — not just one of them. Multiplying both top and bottom by the same quantity is equivalent to multiplying by 1, which preserves the value.

Where this leads. The definition of the derivative involves the complex fraction $\frac{f(x+h) - f(x)}{h}$; after substituting, you often get complex fractions that must be simplified before taking $h \rightarrow 0$. Problem 46 in Worksheet 1A previews this structure.

Explore further. Continued fractions are nested complex fractions extended infinitely: e.g. $1 + \frac{1}{1 + \frac{1}{1 + \dots}}$ = ϕ (the golden ratio). Every rational number has

a finite continued fraction, and irrationals like $\sqrt{2}$ and e have beautiful periodic or patterned ones. See Hardy & Wright, *An Introduction to the Theory of Numbers*, Ch. 10.

$$27. \quad \frac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}}$$

LCD of inner fractions. xy .

Multiply top and bottom by xy . $\frac{xy \cdot \frac{1}{x} + xy \cdot \frac{1}{y}}{xy \cdot \frac{1}{x} - xy \cdot \frac{1}{y}} = \frac{y + x}{y - x}$.

Note: Equivalently write as $\frac{x+y}{y-x}$; note this equals $-\frac{x+y}{x-y}$.

$$28. \quad \frac{\frac{2}{x+1} - \frac{1}{x}}{\frac{3}{x}}$$

LCD of inner fractions. $x(x+1)$.

Numerator *imes* $x(x+1)$. $2x - (x+1) = x - 1$.

Denominator *imes* $x(x+1)$. $3(x+1)$.

Result. $\frac{x-1}{3(x+1)}$.

Note: Verify at $x = 2$: $(\frac{2}{3} - \frac{1}{2}) / \frac{3}{2} = \frac{1}{6} \cdot \frac{2}{3} = \frac{1}{9}$; formula gives $\frac{1}{9}$. ✓

29. $\frac{1 - \frac{1}{x^2}}{1 + \frac{1}{x}}$

LCD of inner fractions. x^2 .

Multiply top and bottom by x^2 . $\frac{x^2 - 1}{x^2 + x}$.

Factor. $\frac{(x - 1)(x + 1)}{x(x + 1)}$.

Cancel. $\frac{x - 1}{x}$.

Note: Restriction: $x \neq 0$, $x \neq -1$.

30. $\frac{\frac{a}{b} + \frac{b}{a}}{\frac{a^2 - b^2}{ab}}$

Numerator: combine over ab . $\frac{a^2 + b^2}{ab}$.

Now divide. $\frac{a^2 + b^2}{ab} \div \frac{a^2 - b^2}{ab} = \frac{a^2 + b^2}{ab} \cdot \frac{ab}{a^2 - b^2} = \frac{a^2 + b^2}{a^2 - b^2}$.

Note: The numerator $a^2 + b^2$ and denominator $a^2 - b^2 = (a - b)(a + b)$ share no common factors (assuming $a \neq \pm b$).

Part E: Reasoning About Fraction Behaviour (Problems 31–36)

Concept. This part builds the intuition underlying **limits**: what value does an expression approach as a variable grows without bound, or shrinks toward zero? You do not yet need formal limit notation — but you are doing exactly the same reasoning. The key insight: *a fraction approaches a value without necessarily reaching it.*

Method. For large x : the highest-power terms dominate. For $x \rightarrow 0$: a very small denominator makes the fraction very large; a very small numerator makes it approach 0. Rewriting as a sum (e.g. $\frac{x+1}{x} = 1 + \frac{1}{x}$) often reveals the limiting value immediately.

Watch out. Confusing “approaches 0 with “equals 0, or “approaches 1 with “equals 1. An expression can get as close to a value as you like without ever touching it. This is why calculus needs the concept of a *limit* rather than just substitution.

Where this leads. The calculus notation $\lim_{x \rightarrow \infty} f(x) = L$ makes this precise. Horizontal asymptotes of rational functions (e.g. $y = 1$ for $y = \frac{x}{x+1}$) are exactly these limiting values. Problems 34 and 36 preview the concept of an asymptote.

Explore further. These “approaching without reaching questions connect to **Zeno’s paradoxes** (Achilles and the tortoise). The resolution lies in the mathematics of infinite series, which Part B of Worksheet 1A begins to explore.

31. Consider $\frac{n}{10}$ where $n > 0$.

(a) As $n \rightarrow \infty$: $\frac{n}{10} \rightarrow \infty$. Each extra unit of n adds $\frac{1}{10}$ to the fraction; there is no upper limit.

(b) As $n \rightarrow 0^+$: $\frac{n}{10} \rightarrow 0$. A tiny numerator over a fixed denominator gives a tiny fraction.

Note: The fraction is proportional to n — linear growth in the numerator means linear growth in the value.

32. Consider $\frac{10}{n}$ where $n > 0$.

(a) As $n \rightarrow \infty$: $\frac{10}{n} \rightarrow 0$. Dividing a fixed quantity into more and more pieces makes each piece smaller.

(b) As $n \rightarrow 0^+$: $\frac{10}{n} \rightarrow \infty$. Dividing by a near-zero number gives a huge result.

Note: $\frac{10}{n}$ and $\frac{n}{10}$ are reciprocals. When one grows, the other shrinks — and vice versa.

33. For $\frac{x+1}{x}$ where $x > 0$:

(a) **Rewrite.** $\frac{x+1}{x} = \frac{x}{x} + \frac{1}{x} = 1 + \frac{1}{x}$.

(b) **As $x \rightarrow \infty$.** $\frac{1}{x} \rightarrow 0$, so $1 + \frac{1}{x} \rightarrow 1$.

(c) **Why.** $1/x$ becomes negligibly small compared to 1 as x grows. The fraction approaches 1 from above but never reaches it.

Look ahead: The function $y = 1 + 1/x$ has the horizontal asymptote $y = 1$ — its graph gets closer and closer to $y = 1$ but never touches it.

34. $\frac{x^2}{x^2+1}$ for positive x :

(a) $x = 1$: $\frac{1}{2} = 0.5$; $x = 10$: $\frac{100}{101} \approx 0.990$; $x = 100$: $\frac{10000}{10001} \approx 0.9999$.

(b) As $x \rightarrow \infty$: approaches 1. Divide top and bottom by x^2 : $\frac{1}{1 + 1/x^2} \rightarrow \frac{1}{1} = 1$.

(c) Never reaches 1: the denominator $x^2 + 1$ is always strictly greater than the numerator x^2 .

Note: The horizontal asymptote is $y = 1$.

35. Fraction $\frac{a}{b}$ with $a, b > 0$:

(a) **Double both.** $\frac{2a}{2b} = \frac{a}{b}$. Unchanged — multiplying by $\frac{2}{2} = 1$.

(b) **Add 1 to both.** Test: $\frac{1}{2} \rightarrow \frac{2}{3}$ (increased); $\frac{3}{4} \rightarrow \frac{4}{5}$ (increased); but $\frac{3}{2} \rightarrow \frac{4}{3}$ (decreased).

(c) **Pattern.** If $a/b < 1$ (numerator smaller), adding 1 to both *increases* the value (moves it toward 1). If $a/b > 1$ (numerator larger), adding 1 *decreases* it (also moves it toward 1). Adding the same positive number to both top and bottom always pulls the fraction toward 1.

Note: This is why fractions of similar numbers cluster near 1: $\frac{99}{100}$ and $\frac{100}{101}$ are both very close to 1.

36. $\frac{1}{1 + \frac{1}{x}}:$

(a) **Simplify.** Multiply top and bottom by x : $\frac{x}{x+1}$.

(b) **As $x \rightarrow \infty$.** $\frac{x}{x+1} = \frac{1}{1+1/x} \rightarrow 1$.

(c) **As $x \rightarrow 0^+$.** $\frac{x}{x+1} \rightarrow \frac{0}{1} = 0$.

Look ahead: Compare with problem 33: $\frac{x}{x+1}$ and $\frac{x+1}{x}$ are reciprocals. One approaches 1 from below; the other from above.

Part F: Multi-Step Problems (Problems 37–40)

Concept. These problems combine factoring, cancellation, and fraction arithmetic in sequence. The art is in choosing the right order of operations: *factor everything first, cancel what you can, then perform the remaining arithmetic*. Rushing to expand brackets before cancelling creates unnecessary complexity.

Method. For products and quotients: factor all numerators and denominators before doing anything else. For sums/differences: find the LCD only after identifying the simplest form of each term. For equations: multiply through by the LCD to clear all fractions, then solve the resulting polynomial equation.

Watch out. When you multiply both sides of an equation by an expression containing a variable, you must check that the variable value you find does not make that expression zero. If it does, it is an *extraneous solution* and must be discarded.

Where this leads. Problem 40 (the sum of a number and its reciprocal) is a classic quadratic hidden inside a fraction equation. This technique — clearing fractions to obtain a polynomial — is used routinely in solving differential equations and optimisation problems.

Explore further. Problem 39 gives the **harmonic mean**: $H = \frac{2ab}{a+b}$ (taking $c = ab/(a+b)$ and doubling). The harmonic mean satisfies $H \leq G \leq A$ (harmonic $\leq$ geometric $\leq$ arithmetic), a famous inequality in analysis. It appears in physics (parallel resistances, average speeds over equal distances).

37. Simplify: $\frac{x^2 - 1}{x^2 + 2x} \cdot \frac{x^2 + 4x + 4}{x^2 + x - 2} \div \frac{x + 2}{x}$

Factor everything. $x^2 - 1 = (x - 1)(x + 1)$; $x^2 + 2x = x(x + 2)$; $x^2 + 4x + 4 = (x + 2)^2$; $x^2 + x - 2 = (x + 2)(x - 1)$.

Rewrite as single fraction. $\frac{(x - 1)(x + 1)}{x(x + 2)} \cdot \frac{(x + 2)^2}{(x + 2)(x - 1)} \cdot \frac{x}{x + 2}$.

Cancel. $(x - 1)$ cancels; x cancels; three factors of $(x + 2)$ cancel with the three in the denominator. Result: $\frac{x + 1}{x + 2}$.

Note: Keep track of how many times each factor appears in numerator vs denominator.

38. Express as a single fraction: $\frac{1}{x} + \frac{1}{x + 1} + \frac{1}{x + 2}$

LCD. $x(x + 1)(x + 2)$.

Rewrite each term. $\frac{(x + 1)(x + 2) + x(x + 2) + x(x + 1)}{x(x + 1)(x + 2)}$.

Expand the numerator. $(x^2 + 3x + 2) + (x^2 + 2x) + (x^2 + x) = 3x^2 + 6x + 2$.

Result. $\frac{3x^2 + 6x + 2}{x(x + 1)(x + 2)}$.

Note: The numerator does not factor nicely (discriminant = $36 - 24 = 12$, not a perfect square), so this is the simplest form.

39. If $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$, solve for c .

Combine the LHS. $\frac{a + b}{ab} = \frac{1}{c}$.

Take reciprocals. $c = \frac{ab}{a + b}$.

Note: This formula gives the combined focal length of two lenses, the total resistance of two parallel resistors, and the harmonic mean $H(a, b) = \frac{2ab}{a + b}$ when doubled.

40. The sum of a number and its reciprocal is $\frac{5}{2}$.

(a) **Write the equation.** $x + \frac{1}{x} = \frac{5}{2}$.

(b) **Clear fractions (multiply by $2x$).** $2x^2 + 2 = 5x$, i.e. $2x^2 - 5x + 2 = 0$.

(c) **Solve.** Factor: $(2x - 1)(x - 2) = 0$, so $x = \frac{1}{2}$ or $x = 2$.

Check. $\frac{1}{2} + 2 = \frac{5}{2}\checkmark$; $2 + \frac{1}{2} = \frac{5}{2}\checkmark$. Both solutions are valid (and they are reciprocals of each other!).

Look ahead: Any equation of the form $x + 1/x = k$ reduces to a quadratic. Numbers that equal their own reciprocal satisfy $x = \pm 1$.